

# Products – Control (PID Tuning)

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**Menu: Products > Control**

Configures the PID controller that adjusts the motor or valve to maintain the target rate.

**Primary Settings**

| Setting      | Description                                                                                                           |
|--------------|-----------------------------------------------------------------------------------------------------------------------|
| 1. Gain      | Rate adjustment factor. Higher product volume requires lower gain. The effect is exponential — adjust very carefully. |
| 2. Integral  | Uses accumulated error to reach target rate faster                                                                    |
| 3. Max Power | Maximum PWM power delivered to motor or valve                                                                         |
| 4. Min Power | Minimum PWM power delivered to motor or valve                                                                         |

**Adjustment Process**

1. **Initial Setup** — Set minimum power to the minimum needed to move the motor/valve. Set maximum power to 100. Set Integral to 0.
2. **Gain Adjustment** — Slowly increase gain until the system overshoots the target, then reduce slightly to stabilize.
3. **Integral** — If needed, increase integral to reach the target rate more quickly.

**Optional / Advanced Settings**

| Setting                      | Description                                       |
|------------------------------|---------------------------------------------------|
| 1. Deadband                  | Error % below which no adjustment is made         |
| 2. BrakePoint                | Error % where adjustment switches to slow rate    |
| 3. Slow Adjustment           | % of full adjustment to use at slow rate          |
| 4. PID loop time             | Adjustment loop interval in milliseconds          |
| 5. Slew Rate                 | Maximum total PWM change per PID loop             |
| 6. Max Integral change       | Maximum Integral PWM change per PID loop          |
| 7. Min Start % (timed combo) | % of target rate for a faster start from 0        |
| 8. Adjust time (timed combo) | Milliseconds the valve adjusts                    |
| 9. Pause time (timed combo)  | Milliseconds the valve pauses adjustment          |
| 10. Min flow sensor Hz       | Minimum valid flow sensor pulse frequency         |
| 11. Max flow sensor Hz       | Maximum valid flow sensor pulse frequency         |
| 12. Flow sensor sample size  | Number of pulse times used to calculate median Hz |

## Controls

| Control                                                                                                    | Action                                      |
|------------------------------------------------------------------------------------------------------------|---------------------------------------------|
|  Navigate left           | Previous product                            |
|  Navigate right         | Next product                                |
|  Load defaults          | Restore default PID values                  |
|  Open rate graph        | Display the rate graph                      |
|  Gain +                 | Increase Gain value                         |
|  Gain –                 | Decrease Gain value                         |
|  Integral +             | Increase Integral value                     |
|  Integral –             | Decrease Integral value                     |
|  Max Power +            | Increase Max Power value                    |
|  Max Power –            | Decrease Max Power value                    |
|  Min Power +            | Increase Min Power value                    |
|  Min Power –            | Decrease Min Power value                    |
|  Show advanced settings | Expand the advanced/optional settings panel |
|  Cancel                 | Discard changes                             |
|  Save                   | Save settings                               |